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Experiment 14: Introduction to Deluge Scripting: Writing Custom Functions and Basic Programs

1. Aim

To learn the basics of Deluge, the scripting language used in Zoho Creator, by writing simple programs and custom functions that extend the Student Project Tracker with logic that cannot be set through the visual tools alone.

2. Introduction to Deluge

Deluge is a scripting language designed for Zoho applications. It uses a readable syntax, supports variables, conditions, loops, lists and functions, and can read and update the records of a Zoho Creator form. Scripts can be attached to form events, such as when a record is submitted, or placed inside custom functions that are called from buttons, workflows or other scripts.

Several basic programming concepts appear in every script. A variable stores a value that can change during the script. A conditional statement runs a block of code only when a condition is true. A loop repeats a block of code for each item in a list. A function groups code under a name so that it can be reused. The programs below demonstrate each of these concepts in a Deluge context.

3. Sample Programs

Program 1: Display a welcome message using the info statement. The info statement writes text to the script log of the editor, which is useful for testing.

```
info "Welcome to the Student Project Tracker";
```

Program 2: Store a value in a variable and combine it with text. The program below builds a message showing the student's name and the project title.

```
studentName = input.Student_Name;
projectTitle = input.Project_Title;
info "Proposal from " + studentName + " on: " + projectTitle;
```

Program 3: A conditional check that warns when a proposal is marked Ongoing without a domain selected.

```
if (input.Status == "Ongoing" && input.Domain == null)
{
    info "Please select a domain for this project.";
}
```

Program 4: A loop over a list of statuses, showing the use of for each over a collection.

```
statuses = List("Planned", "Ongoing", "Completed");
for each s in statuses
{
    info "Status option: " + s;
}
```

Program 5: A custom function that returns a short label for a status, showing how a function groups logic and returns a value.

```
string statusLabel(string st)
{
    if (st == "Completed")
    {
        return "Done";
    }
    return "In progress";
}
info statusLabel("Completed");
```

The sample code above is illustrative. Exact function names, date handling and the form field names used in Creator should be checked against the Deluge reference in the Zoho Creator documentation, because details can differ between versions. The field names used here, such as Student_Name and Project_Title, depend on the names given to the fields in the application.

4. Procedure

- The Deluge script editor was opened from the Project Entry form, and each program was typed into a new custom function.
- Each function was run with the Run option, and the output was read in the script log.
- Each program was then tested with several records, including one with a blank Domain, to confirm the condition worked.
- Errors such as missing semicolons and mismatched braces were corrected based on the messages shown by the editor.

5. Observations

- The info statement was useful for debugging, because its output appeared in the log and showed the values held by each variable.
- Conditions and loops behaved as expected when tested with sample records.
- String concatenation with the plus operator required the variables to be text; combining text with a number needed a conversion first.
- Validation written in Deluge can cover rules that the visual form settings cannot express, such as a condition that depends on two fields.

6. Discussion

Deluge scripting takes the application beyond what the visual builder can do. The conditional program shows how a business rule can be enforced by code, and the function shows how logic can be reused in several places, such as in reports and workflows. The exercise also highlights good programming habits: name variables clearly, keep each function short, and test each program with normal, empty and unusual values. The main limitation is that the programs were tested in a single small application, so performance and error handling for large data volumes were not examined.

7. Result and Conclusion

Basic Deluge scripting was learned, covering output, variables, text handling, conditions, loops, lists and custom functions. These skills allow a no-code application to be extended with custom logic. This is the practical step from no-code towards low-code development, and it completes the progression of this series from application setup to automation.